

Pharma*stock*
Stock and care flows

How a medicine, a patient and a procedure move through ERPNext Stock and Frappe Health, step by step, with the rules and the permissions behind each one.

Version 16 (Frappe, ERPNext, Frappe Health) · github.com/Arielpetit/pharmastock · written from the source code

CONTENTS

What this document covers

Each flow page has the same layout: a sequence diagram, then the rules and errors, then what exists afterwards. Numbers in the diagram match the steps in order.

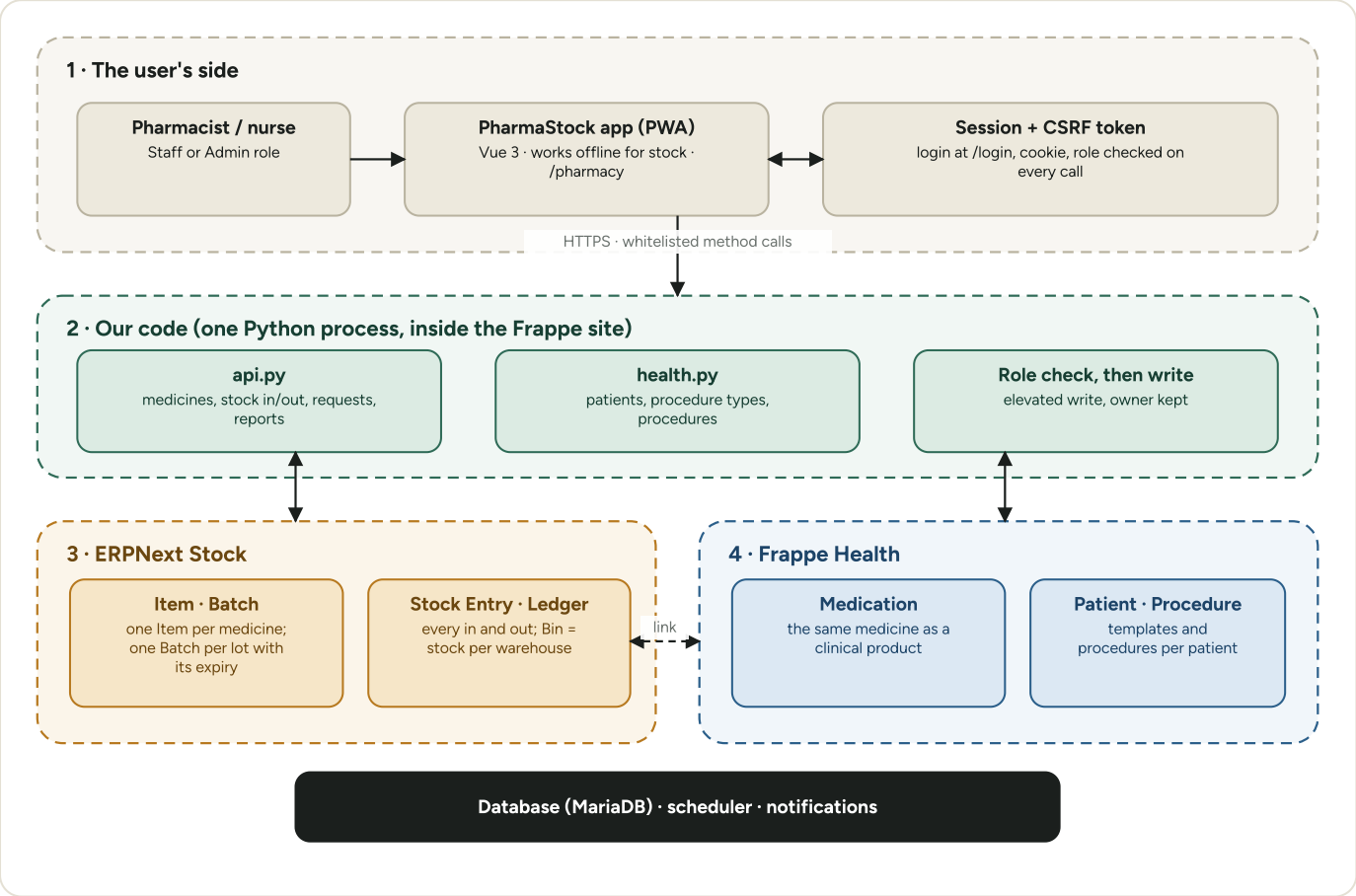
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Colours: App / user · PharmaStock API · ERPNext Stock · Frappe Health

OVERVIEW

The idea in one page

A health center needs two things that normally live apart: the **pharmacy stock** and the **patient care**. PharmaStock keeps both in ERPNext, in two standard modules, and joins them through one rule: **medicine only leaves the stock for a reason, and a patient only receives medicine through a procedure.**



MODULE	OWNS	EXAMPLES
ERPNext Stock	Quantities, batches, expiry, every movement (the ledger)	Item, Batch, Stock Entry, Stock Ledger, Bin, Material Request
Frappe Health	People and care	Patient, Medication, Clinical Procedure Template, Clinical Procedure
PharmaStock	The screens, the rules between the two, the permissions	api.py, health.py, the /pharmacy app

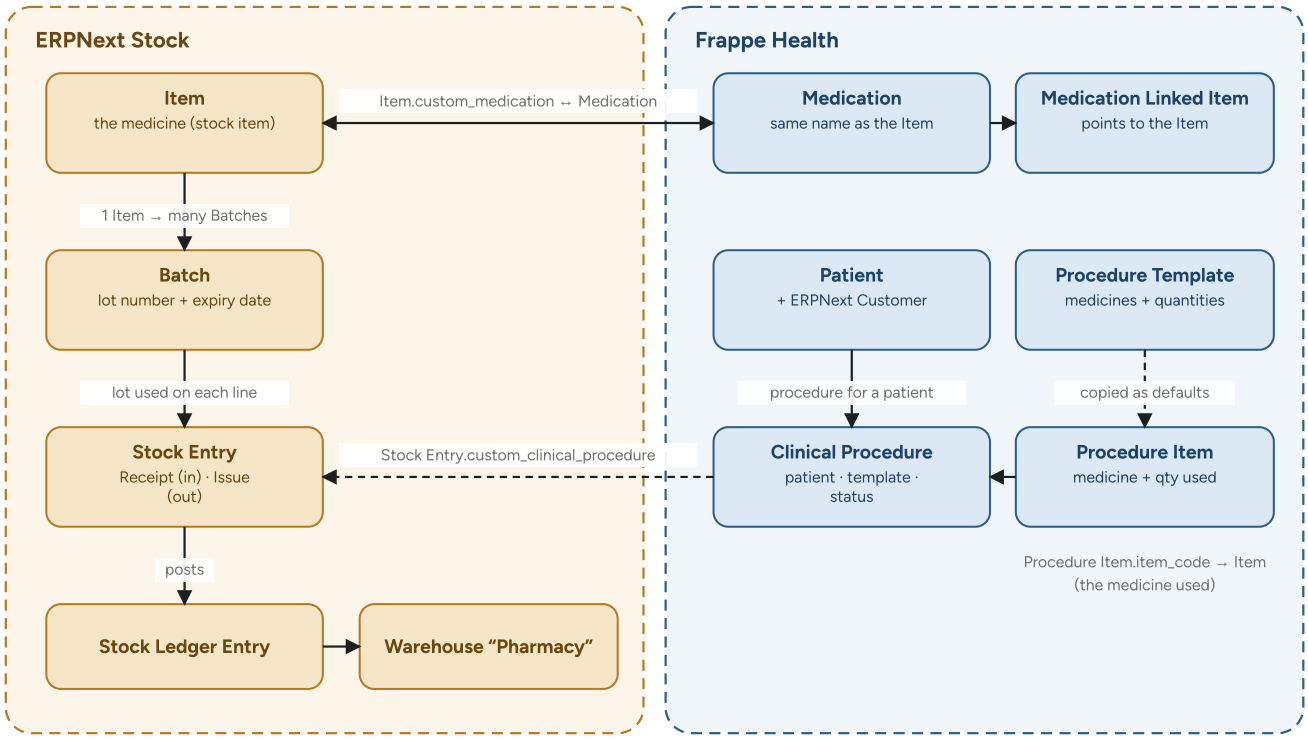
One request, one transaction. Every button is one call to the API. The API writes to Stock and Health together; if any step fails (for example not enough valid stock), nothing is saved. No half-finished records.

How stock reaches a patient. Medicine given to a patient is recorded through a **procedure** (Flow 5). The Remove stock page (Flow 7) covers the reasons that have no patient: damaged, expired, transfer, other adjustment. The older "Dispensed to a patient" reason still exists on that page.

OVERVIEW

How the records are linked

One medicine is **one Item** in Stock and **one Medication** in Health, pointing at each other. A procedure copies its medicines from a template, and the stock it uses is a normal Stock Entry linked back to the procedure.

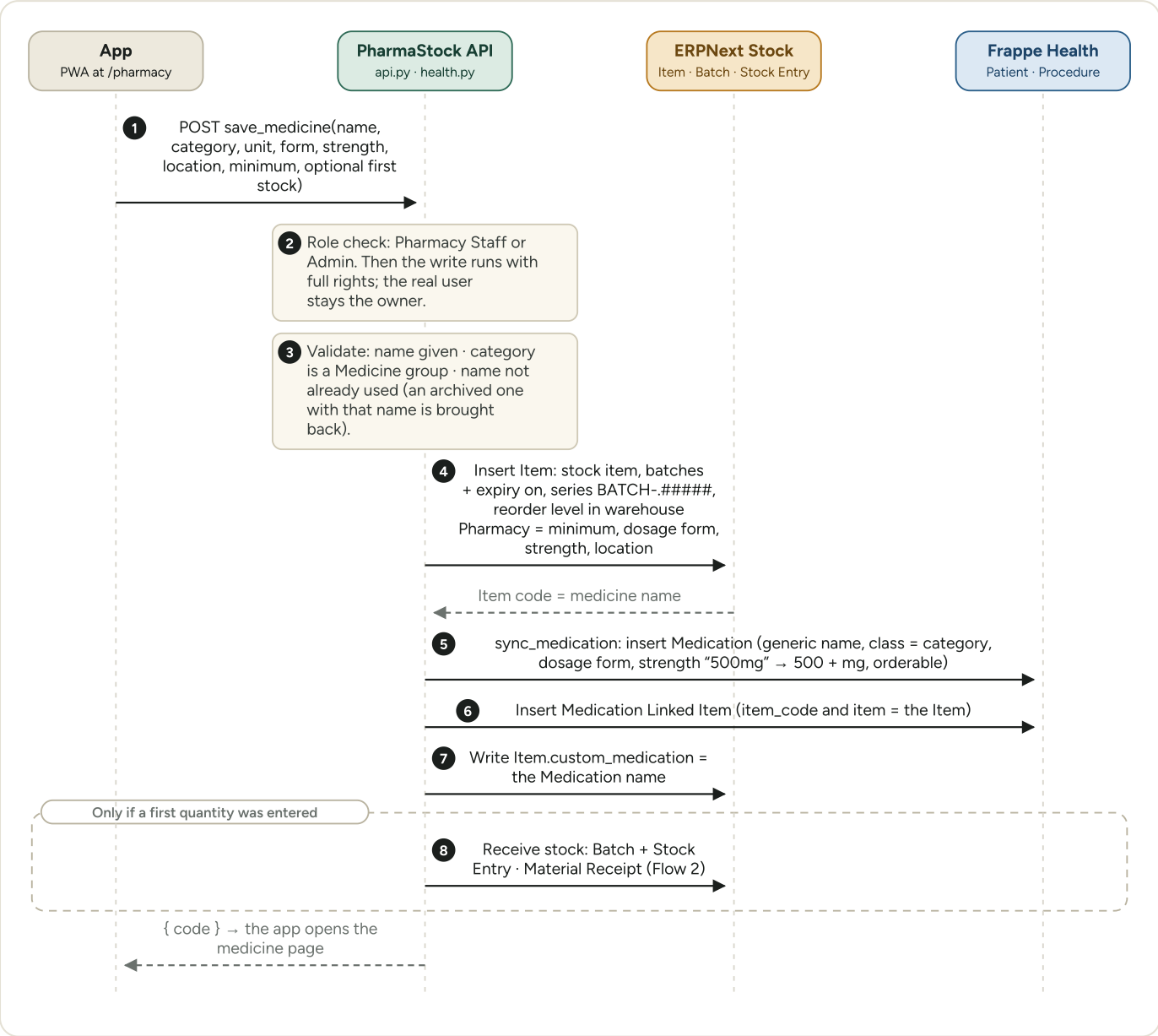


USER ACTION	ERPNEXT STOCK RECORD	FRAPPE HEALTH RECORD
Add a medicine	Item (+ Batch + Stock Entry if first stock)	Medication + Medication Linked Item
Receive a delivery	Batch + Stock Entry · Material Receipt	—
Register a patient	—	Patient (+ Customer in ERPNext)
Create a procedure type	service Item (created by Health)	Clinical Procedure Template + items
Record a procedure	Stock Entry · Material Issue (reason "Used in a procedure")	Clinical Procedure + items
Remove stock (no patient)	Stock Entry · Material Issue	—
Request stock	Material Request	—

FLOW 1

Create a medicine

The pharmacist adds a medicine once. The API creates the Stock side and the Health side together, so they cannot drift apart.



RULES AND ERRORS

- Name is required. The category must be one of the Medicine item groups.
- A name already in use is refused: "already exists. Open it and use Add Stock instead."
- An archived medicine with the same name is restored with the new details.
- The unit ("Counted in") is fixed at creation. Minimum stock sets the reorder level (reorder quantity = twice the minimum).
- Strength "500mg" becomes 500 mg in Health; text that does not parse stays on the Item and is stored as 0 mg.

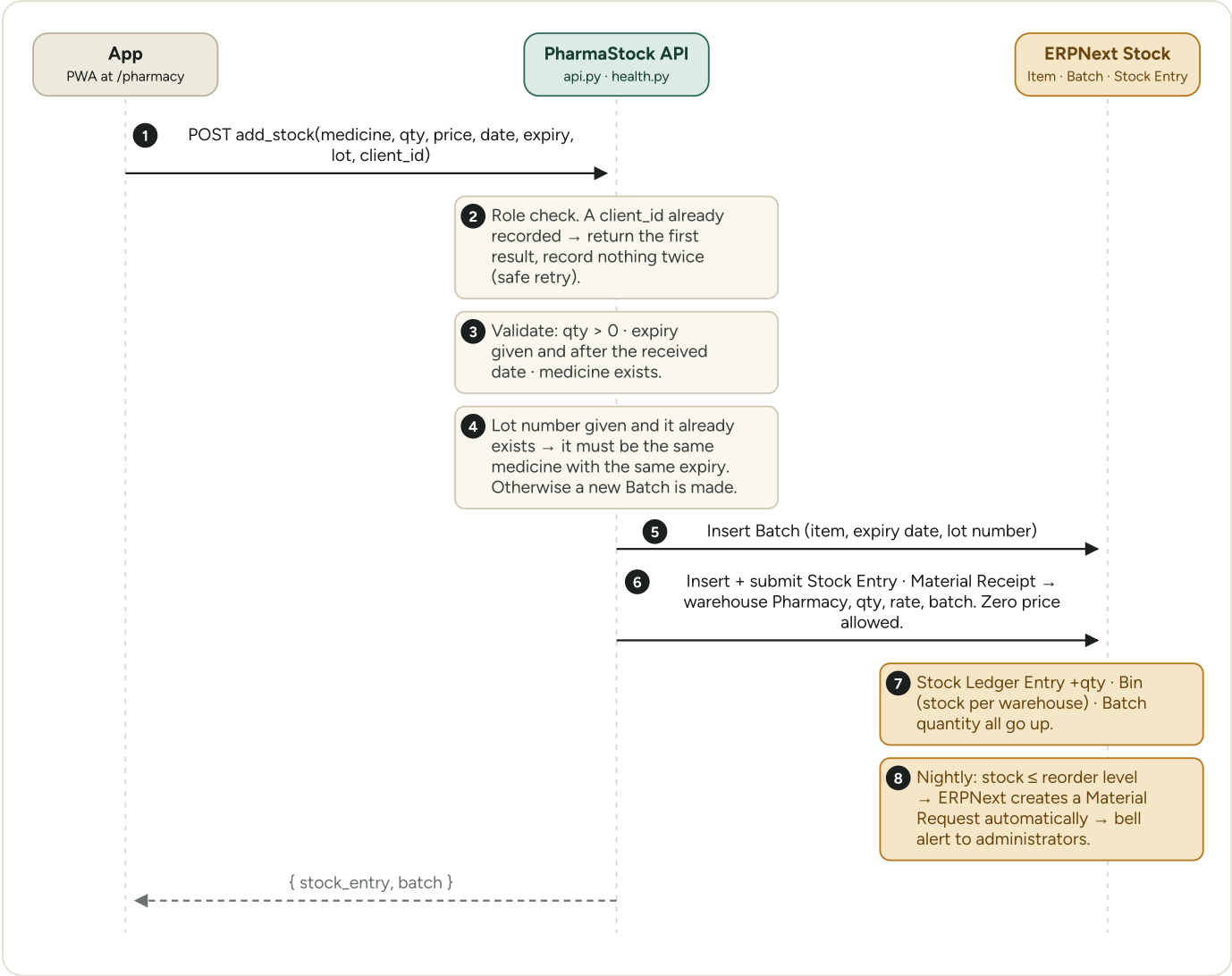
WHAT EXISTS AFTERWARDS

- Item: stock item with batch and expiry tracking, reorder level, owner = the user.
- Medication with the same name, linked both ways to the Item.
- Optionally a Batch and a Stock Entry · Material Receipt for the first stock.

FLOW 2

Receive stock

Every delivery is recorded with an expiry date. Stock is tracked per batch, so the oldest valid batch can always be used first.



RULES AND ERRORS

- Quantity must be above 0. The expiry date is required and must be after the date received.
- A lot number that exists must belong to the same medicine and have the same expiry date, otherwise it is refused.
- Price can be 0 (donations); stock is then valued at zero.
- Sending the same client_id twice records one delivery only.
- Low stock is not checked here: ERPNext's nightly job creates the reorder request.

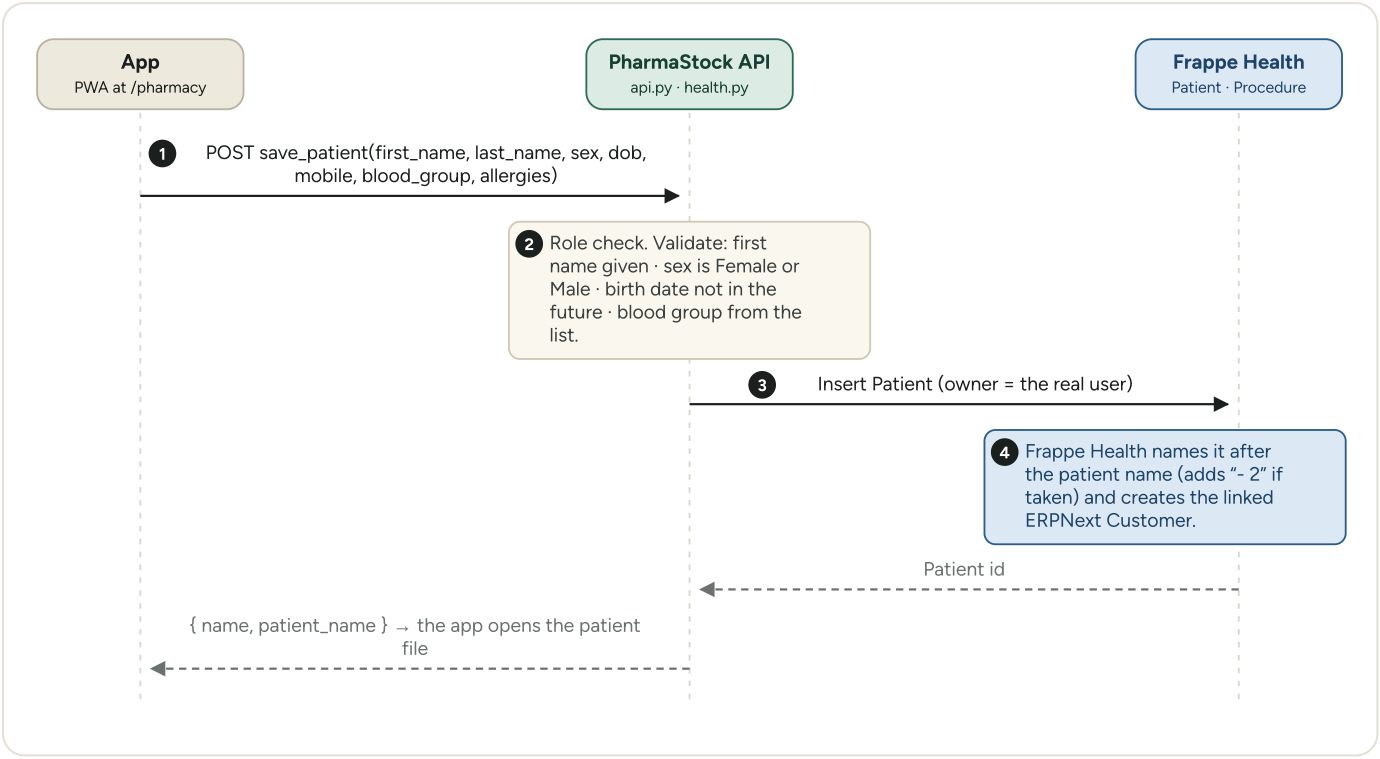
WHAT EXISTS AFTERWARDS

- A Batch (if new) with its expiry date.
- A submitted Stock Entry · Material Receipt, owner = the user.
- Stock Ledger Entry +qty; Bin and Batch quantities updated.

FLOW 3

Register a patient

Patients are Frappe Health records. The pharmacy app only asks for what a clinic needs at the desk.



RULES AND ERRORS

- First name and sex are required. Sex is Female or Male.
- Date of birth cannot be in the future. Blood group must be from the list.
- Two patients with the same name get "- 2", "- 3"… as their id. Search by name or phone in the app.
- Patients are never deleted from the app. The list shows active patients.

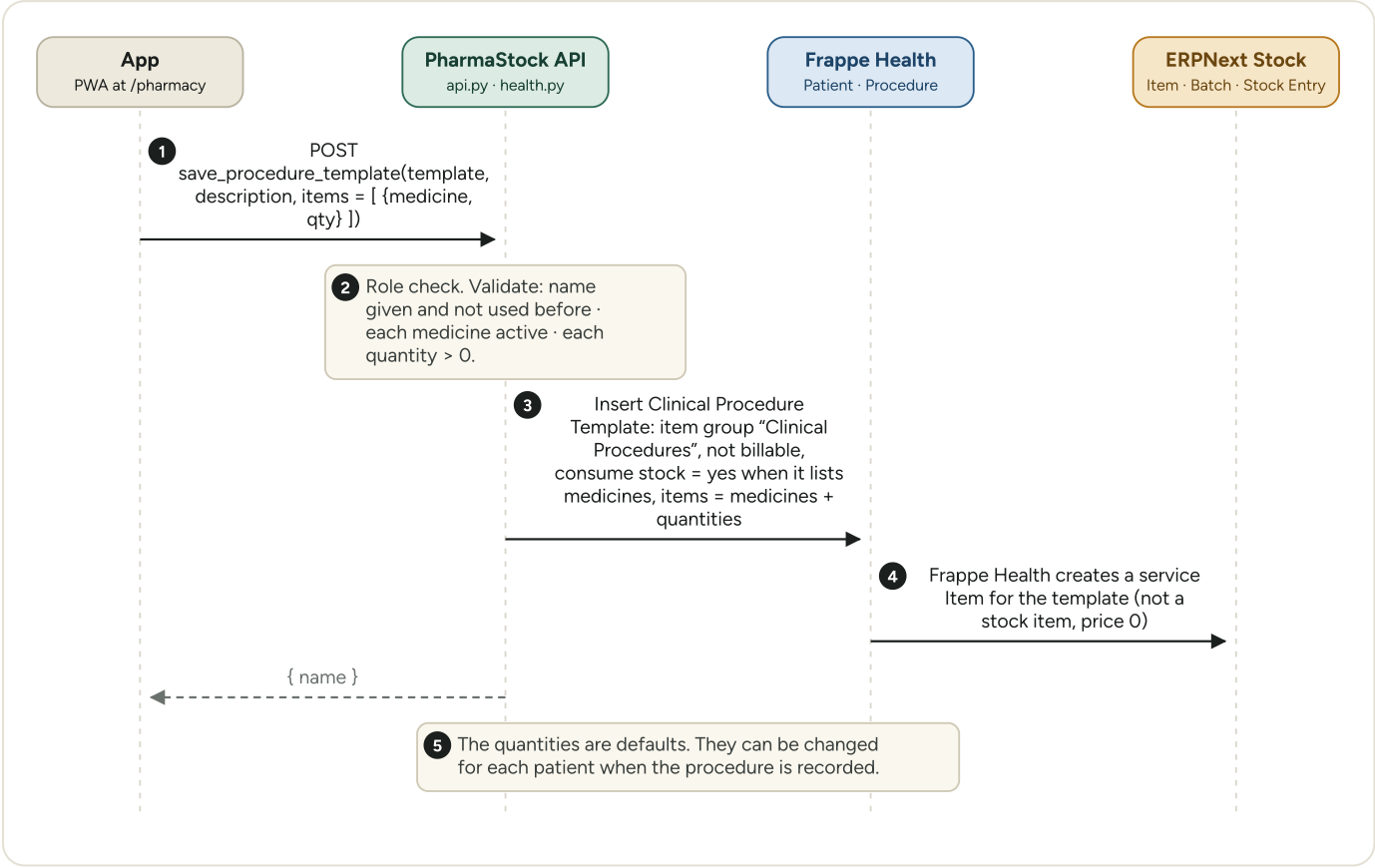
WHAT EXISTS AFTERWARDS

- Patient (Frappe Health), owner = the user.
- A linked ERPNext Customer, created by Frappe Health.
- Age is calculated, never stored.

FLOW 4

Create a procedure type

A procedure type is a care you give often (wound dressing, injection) and the medicines it uses each time.



RULES AND ERRORS

- Name required and unique. Medicines must be active; quantities above 0.
- A type can use no medicine at all (consume stock = no).
- A type can be hidden ("no longer used"). It stays in old records and can be shown again.
- Editing a type changes future procedures only; recorded procedures keep their own medicines.

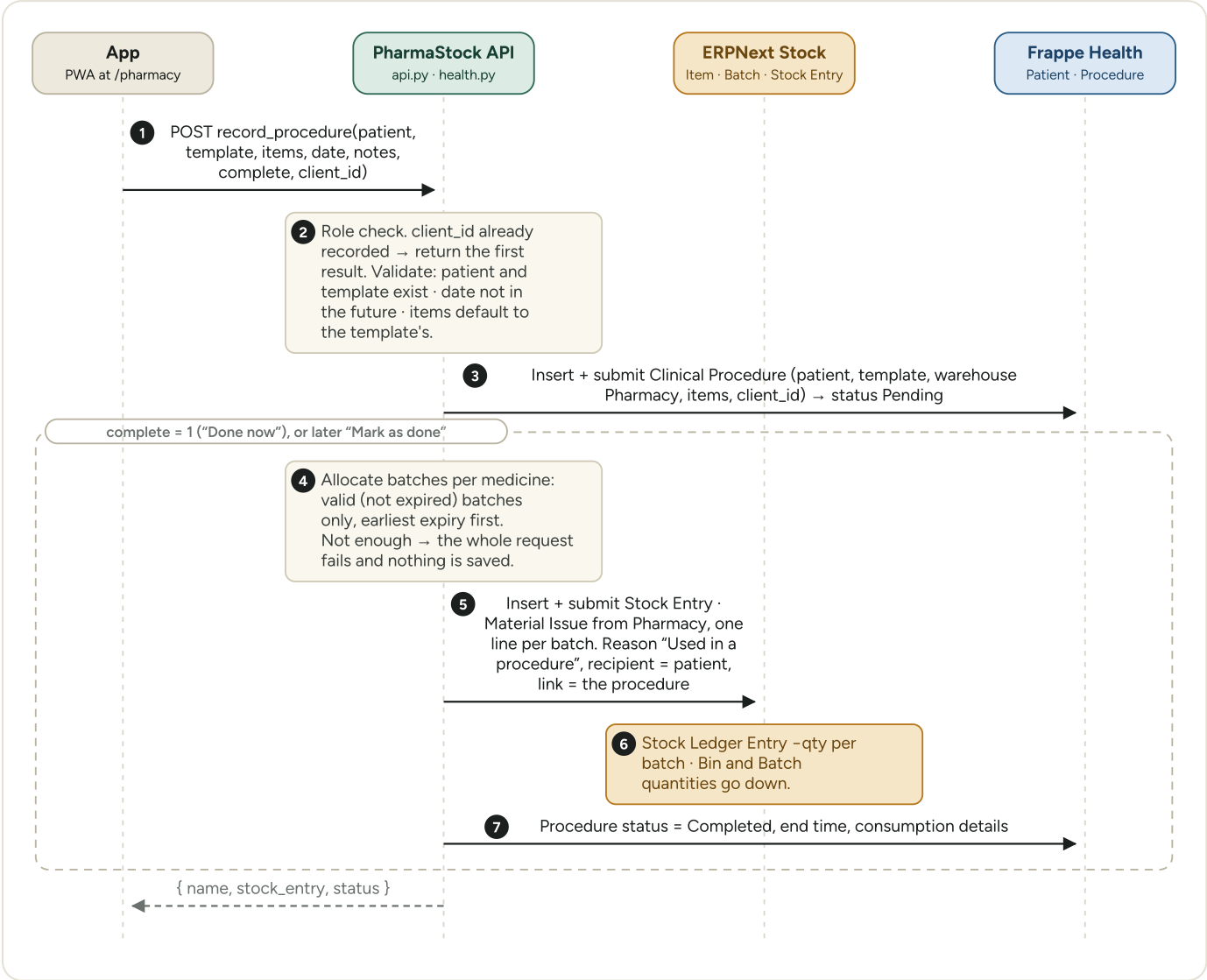
WHAT EXISTS AFTERWARDS

- Clinical Procedure Template with its items (Frappe Health).
- A service Item named like the template (created by Health, not a stock item).
- Item group "Clinical Procedures".

FLOW 5

Record a procedure

This is how medicine reaches a patient. The procedure and the stock movement are created together and linked.



RULES AND ERRORS

- Date cannot be in the future. Medicines default to the template; the nurse can change quantities or add medicines.
- "Done now": stock leaves immediately. "Planned": the procedure waits (status Pending) and nothing leaves until "Mark as done".
- Only valid batches are used, earliest expiry first. Expired batches are never used here.
- Not enough valid stock for any medicine → refused, nothing saved.
- The stock movement shows "Used in a procedure" with the patient's name; the same client_id twice records one procedure.

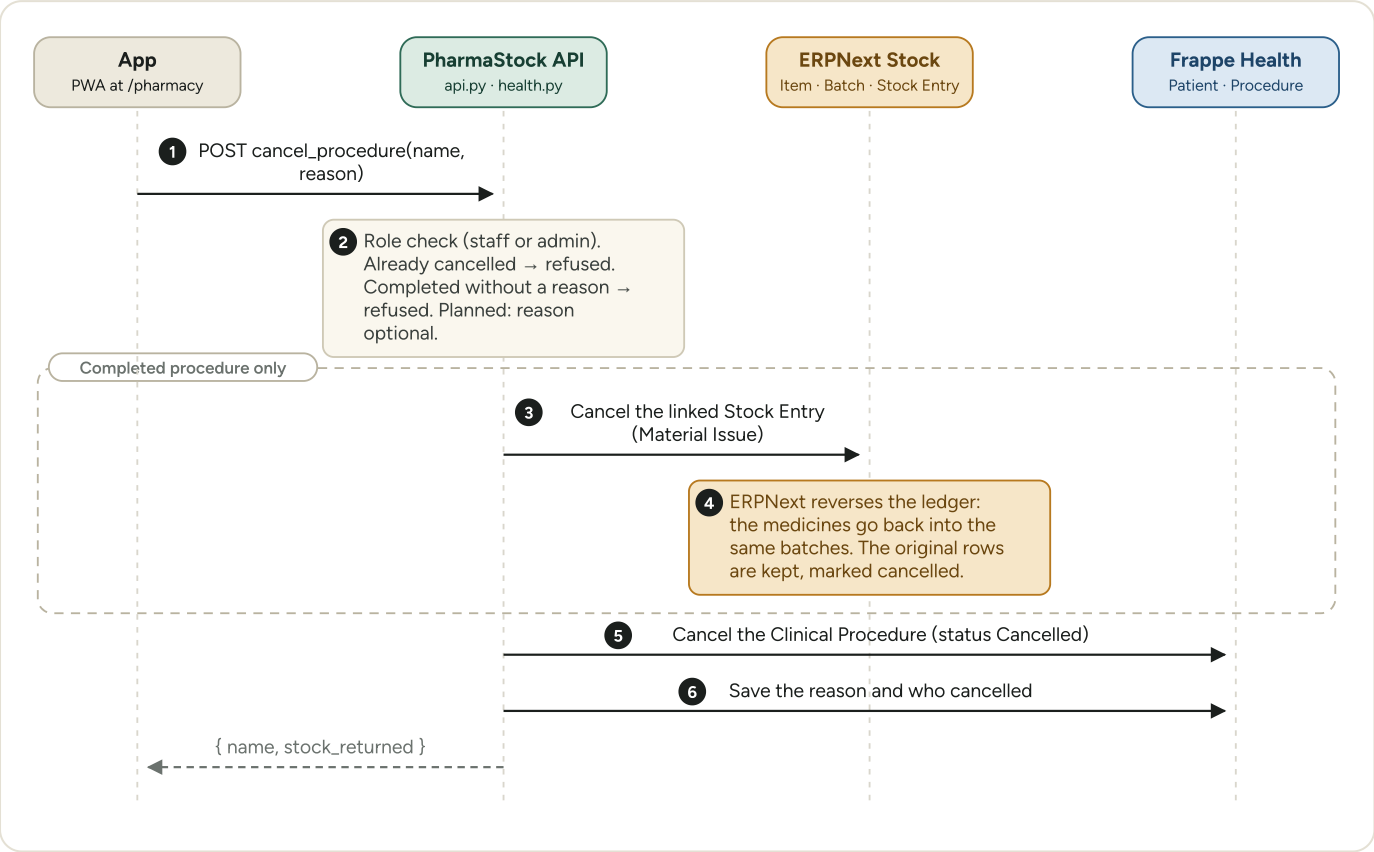
WHAT EXISTS AFTERWARDS

- Clinical Procedure (submitted) with items, status Completed or Pending.
- Stock Entry · Material Issue, linked to the procedure, one line per batch used.
- The patient file lists the procedure; reports list the movement.

FLOW 6

Cancel a procedure

A procedure recorded by mistake can be cancelled by staff or an administrator. Stock comes back; nothing is erased.



RULES AND ERRORS

- A completed procedure needs a reason (for example “Recorded for the wrong patient”).
- A planned procedure can be cancelled without a reason; no stock is involved.
- A procedure can be cancelled once. It cannot be reopened; record a new one instead.
- The medicines return to the same batches they came from, with their original expiry dates.

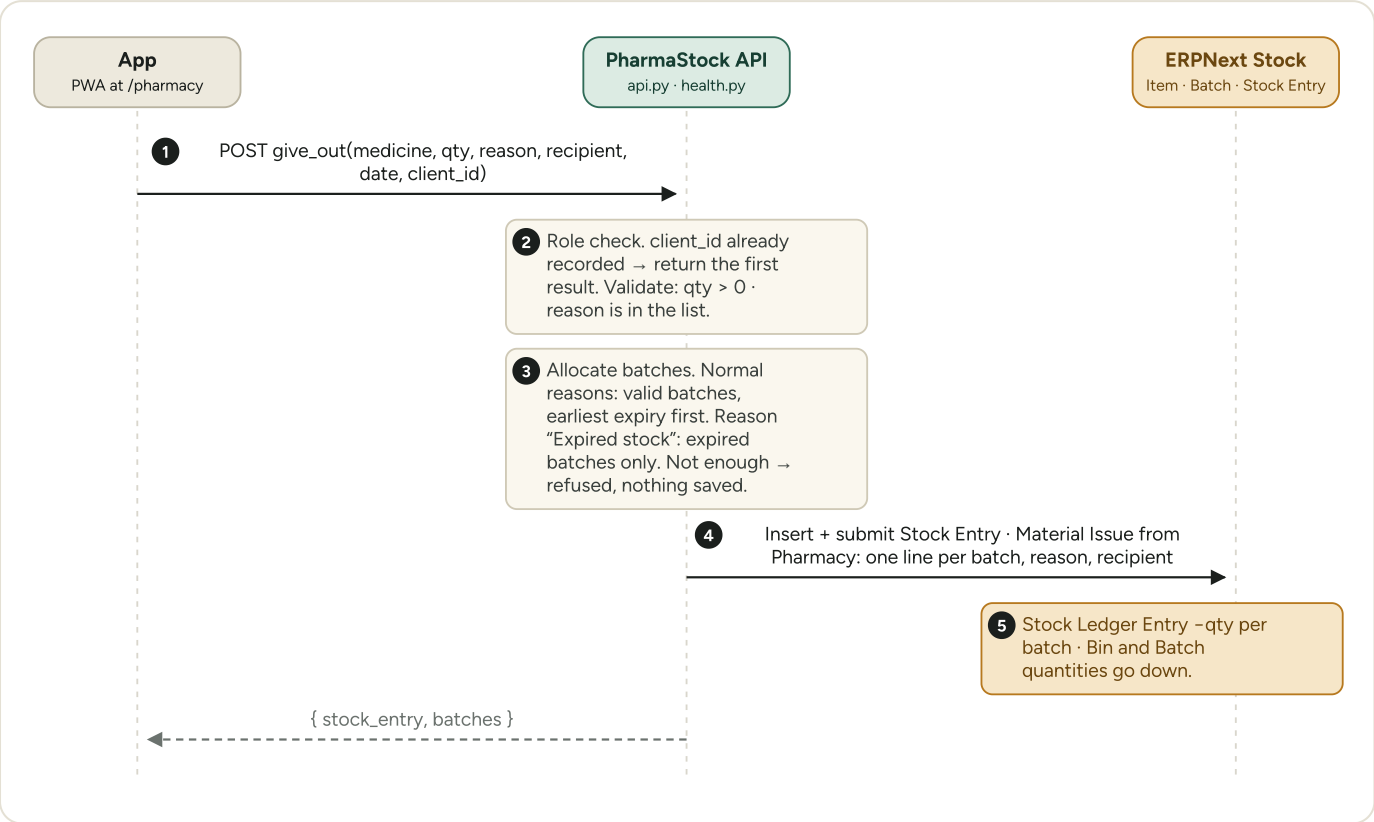
WHAT EXISTS AFTERWARDS

- Stock Entry cancelled: ledger reversed, stock back up.
- Reports, medicine history and dashboard still show the original movement, marked Cancelled, quantity struck through.
- The procedure stays in the patient file as Cancelled with the reason and who cancelled it.

FLOW 7

Remove stock without a patient

Some stock leaves for reasons that have no patient. The Remove stock page records them, with a reason, against the oldest valid batch.



RULES AND ERRORS

- Quantity above 0. A reason is required from the list.
- "Expired stock" uses expired batches only; every other reason uses valid batches only.
- A medicine with only expired stock cannot be given out except as "Expired stock".
- The recipient field is free text (patient name or file number, store name…).

WHAT EXISTS AFTERWARDS

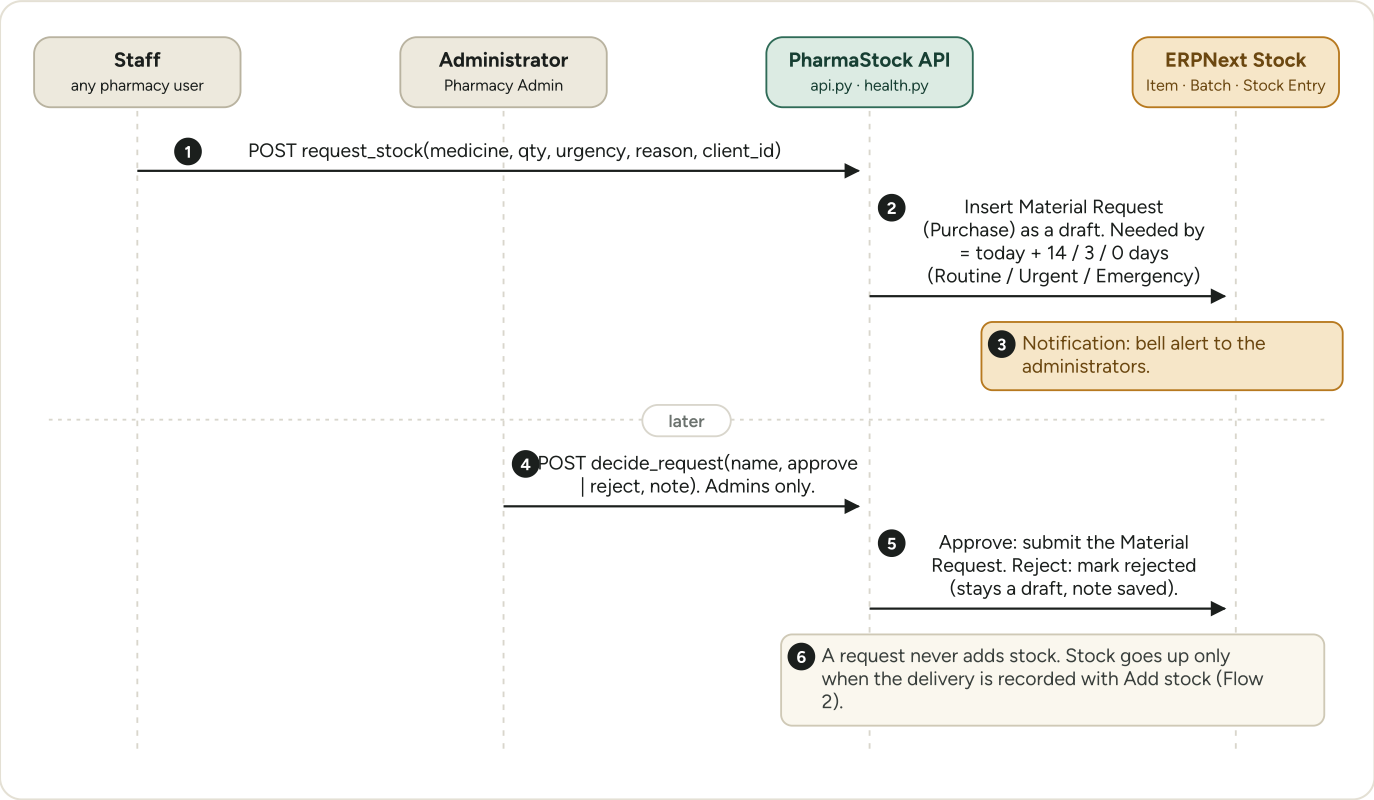
- Stock Entry · Material Issue with reason and recipient.
- Ledger −qty per batch.
- Visible in Reports and in the medicine history.

REASON	BATCHES USED	HAS A PATIENT?
Used in a procedure (Flow 5)	valid, earliest expiry first	yes, through the procedure
Dispensed to a patient	valid, earliest expiry first	free text name
Transfer to another store or department	valid, earliest expiry first	no
Damaged stock	valid, earliest expiry first	no
Expired stock	expired only	no
Other authorized adjustment	valid, earliest expiry first	no

FLOW 8

Request stock

Staff ask for medicine to be ordered. An administrator decides. The request never changes the stock.



RULES AND ERRORS

- Urgency sets the needed-by date: Routine +14 days, Urgent +3, Emergency today.
- Only an administrator can approve or reject. A request can be decided once.
- Approve submits the Material Request; reject keeps it as a draft marked rejected, with the note.
- ERPNext also creates a request by itself when stock falls to the minimum (reorder level).

WHAT EXISTS AFTERWARDS

- Material Request (draft, then submitted if approved).
- Bell notification to the administrators.
- The delivery itself is recorded later with Receive stock (Flow 2).

MAINTENANCE

Keeping medicines in sync

Stock is the source of truth; Health follows. A pharmacy change on a medicine updates its Medication in the same request.

ACTION IN THE APP	STOCK (ITEM)	HEALTH (MEDICATION)
Create	Item created	Medication + Linked Item created, Item.custom_medication set
Edit name, category, form, strength	Item updated	Medication updated (class, dosage form, strength + unit, name field). Its id is not renamed.
Change the minimum or the location	Item reorder level / location	no change (not clinical data)
Delete, never used	Item deleted	Medication and Linked Item deleted
Delete, has history or in a procedure	Item archived (disabled); ledger kept	Medication disabled
Add the same name again	Archived Item restored	Medication re-enabled and updated
Delete while stock remains	refused: give it out or record it as expired first	no change

> **Direction.** The sync is one way, from the pharmacy app into Health. A Medication edited by hand in the ERPNext desk does not change the Item. Existing medicines are synchronised when the app is installed or updated (`bench migrate`).

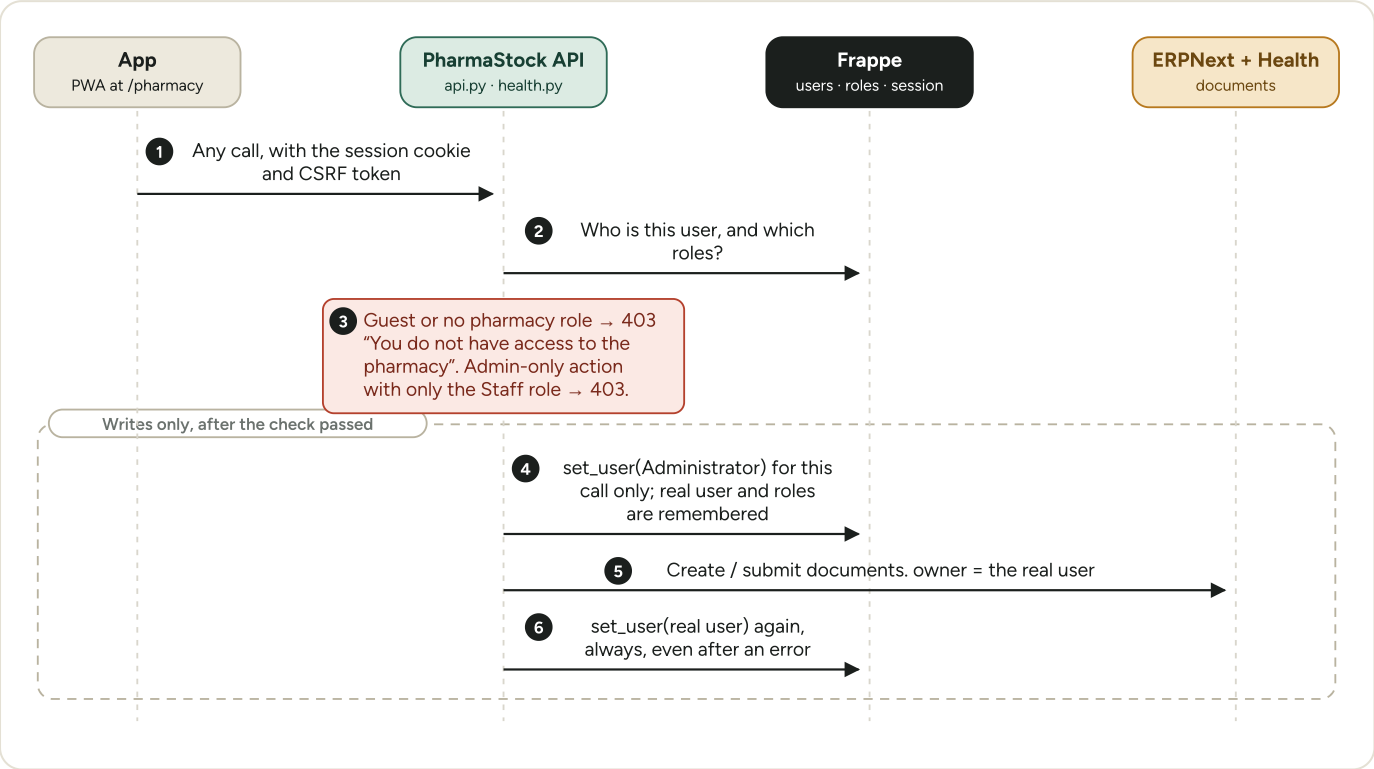
REMINDER: HOW A MEDICINE IS IDENTIFIED

WHERE	IDENTIFIER
Item	item code = the medicine name (e.g. “Paracetamol 500mg Tablet”)
Medication	same string; “name (form)” if a Medication of that name already existed
Batch	BATCH-00001… or the lot number typed by the pharmacist

SECURITY

Permissions

Two pharmacy roles use the app. Neither has access to the ERPNext desk. The API checks the role on every call, then does the work for them with full rights, so nobody needs permissions on ERPNext documents.



ACTION	PHARMACY STAFF	PHARMACY ADMIN
Sign in, dashboard, medicine list, reports, notifications	yes	yes
Add, edit, delete a medicine	yes	yes
Receive stock · Remove stock · Request stock	yes	yes
Approve or reject a stock request	no	yes
Register and edit patients	yes	yes
Create, edit, hide procedure types	yes	yes
Record, complete, cancel a procedure	yes	yes
Create users, change roles, disable users	no	yes
ERPNext desk (/desk), any ERPNext screen	no	no

HOW IT IS ENFORCED

- Both roles have `desk_access = 0`: the desk answers "Not Permitted".
- Every API method starts with the role check. The screens hide what a role cannot do, but the server decides.
- Only `boot` is open to guests, and it returns nothing without a session.
- Each record keeps the real user as owner; a cancellation also keeps its reason and author.

MANAGING USERS

- The administrator creates users in the Staff page: name, email, role, first password.
- Only the two pharmacy roles can be given there. A user with other ERPNext roles is changed by a System Manager only.
- An administrator cannot remove their own admin access.
- Administrator / System Manager keep the desk for setup and repair.

RELIABILITY

Offline, safe retries, notifications

Health centers lose connection. Stock work continues; care work needs a connection because it needs patients and live stock.

WORKS OFFLINE (QUEUED, SENT LATER)	NEEDS A CONNECTION
Receive stock · Remove stock · Request stock	Register patient · procedure types · record, complete, cancel a procedure
Reading the last loaded lists and dashboard	Approve or reject a request · manage staff · reports with new filters

> **Safe retries.** The app gives every write a unique `client_id`. If the same call arrives twice (a double tap, or an offline queue sent again) the server returns the first result and records nothing new. Stock is never counted twice.

NOTIFICATIONS (BELL)

EVENT	WHO IS TOLD	CREATED BY
A batch expires within 60 days	Staff and Admin	ERPNext notification on Batch
Stock fell to the minimum; a reorder request was created	Admin	ERPNext auto reorder + notification
A stock request is waiting for approval	Admin	Notification on Material Request

BEHAVIOUR WITH EXPIRY

- Every outgoing movement picks valid batches by earliest expiry first.
- Expired stock can only leave as “Expired stock”.
- The dashboard, medicine list and reports flag low, expiring and expired stock.

FOR ADMINISTRATORS

Where everything is in the ERPNext desk

The desk is for the site owner. Everything the app creates is a standard ERPNext or Frappe Health record, so it can be inspected there.

TO SEE	OPEN IN THE DESK	LOOK FOR
Medicines	/desk/item (filter item group: Medicine)	custom dosage form, strength, Medication link
Medications (Health)	/desk/medication	linked items table
Batches and expiry	/desk/batch	quantity and expiry per lot
Every stock movement	/desk/query-report/Stock Ledger	voucher = Stock Entry
Stock Entries	/desk/stock-entry	Reason, Recipient, Clinical Procedure
Patients	/desk/patient	linked Customer
Procedure types	/desk/clinical-procedure-template	items = medicines used
Procedures	/desk/clinical-procedure	status, cancel reason
Stock requests	/desk/material-request	urgency, decision note
Users and roles	/desk/user	roles Pharmacy Staff / Pharmacy Admin
Installed apps	/desk/installed-applications	frappe, erpnext, healthcare, pharmastock

> **Do not edit stock records by hand in the desk.** Create movements through the app so reasons, owners and links stay complete. Cancelling a Stock Entry in the desk without cancelling its procedure leaves a procedure that no longer matches the stock.

OPERATIONS

Setup and limits

WHAT THE INSTALLATION NEEDS

- Frappe, ERPNext and **Frappe Health**, all version 16, on the same bench. PharmaStock lists Health as a required app.
- A company created by the ERPNext setup wizard (country Cameroon, currency XAF). The “Pharmacy” warehouse is created for it.
- On install and on every `bench migrate`: roles, custom fields, medicine categories, units, bell notifications, and a Medication for every existing medicine.
- On Frappe Cloud / Press: add Frappe Health to the bench group, add the GitHub repository as an app, deploy, then install the app on each clinic site.

TEST COVERAGE

TEST	CHECKS	COVERS
tests/api_smoke.py	23	Stock in and out, earliest expiry first, refusals, retries, requests, permissions
tests/health_smoke.py	26	Patient, procedure type, done and planned procedures, stock used, cancelling, history
tests/e2e.mjs	21	Browser: staff and admin journeys, offline, French, phone
tests/care_e2e.mjs	14	Browser: medicine, patient, type, procedures, cancel, dashboard

NOT BUILT YET

- Consultations, practitioners and appointments (Frappe Health has them; they need practitioner setup first).
- Billing and invoices for procedures (types are created as not billable).
- Offline recording of patients and procedures.
- Two-way sync: editing a Medication in the desk does not update the Item.
- Email or push notifications (bell only).